

The Regular Dodecahedron Refutes a Centro-Affine Gamma-Moment Characterization

Automated mathematical discovery packet for arXiv:2103.02994

model: GPT5.6; lane: agent_lane_08

19 July 2026

Abstract

Milman asked whether an origin-symmetric tangential body of a Euclidean ball whose Γ_{-2} and Γ_{-4} bodies are Euclidean must be an ellipsoid. We give a negative answer in dimension three: the regular dodecahedron circumscribed about the unit ball. Its twelve facet normals are the vertices of a regular icosahedron, whose exact second and fourth moments are rotationally invariant. The normalization works out to

$$\Gamma_{-2}K = B_2^3, \quad \Gamma_{-4}K = \left(\frac{5}{3}\right)^{1/4} B_2^3,$$

which is precisely the equality relation in the source's Theorem 4.1(3), yet K is a non-ellipsoidal polytope.

1 Source question

For a convex body $K \subset \mathbb{R}^n$ containing the origin, the source uses

$$\|x\|_{\Gamma_{-p}K}^p = \frac{1}{V(K)} \int_{S^{n-1}} |\langle x, \theta \rangle|^p dS_p(K, \theta), \quad dS_p(K, \theta) = h_K(\theta)^{1-p} dS_K(\theta). \quad (1)$$

Theorem 4.1(3) identifies equality in every direction with the relation

$$\Gamma_{-4}K = \left(\frac{n+2}{3}\right)^{1/4} \Gamma_{-2}K, \quad (2)$$

or, after a linear change of coordinates, with K tangential to a Euclidean ball and both $\Gamma_{-2}K$ and $\Gamma_{-4}K$ Euclidean. Remark 4.2 then asks whether this characterizes ellipsoids among origin-symmetric convex bodies [1].

will necessarily be in S_2 -isotropic position. However, we do not know whether the property stated in assertion [3] above characterizes ellipsoids in the class of origin-symmetric convex bodies $\mathcal{K}_e$. This has to do with the fact that the L^p spherical cosine-transform, while being injective on the space of even measures for $p > 0$ which is not an even integer [67], is not injective for $p \in 2\mathbb{N}$ (since in that case $\{|\langle \cdot, \xi \rangle|^p\}$ span a finite dimensional linear space). Consequently, it is easy to construct examples of bodies $K \in \mathcal{K}_e$ different from Euclidean balls so that $\Gamma_{-2}K$ and $\Gamma_{-4}K$ are themselves Euclidean balls (see e.g. [73] Section 4)). But perhaps in conjunction with the assumption that K is a tangential body of a ball, this already forces K to be a ball itself? We leave this as an interesting open problem.

2 Proof intuition

For a tangential polytope, the measures in (1) are supported on the facet normals. Regularity makes their weights equal. The question therefore becomes a finite moment-design problem: choose a symmetric set of normals whose quadratic and quartic moments are both radial. The twelve icosahedral vertices do exactly this. Their polar is the regular dodecahedron, so the moment design simultaneously supplies the required tangential body.

Theorem 1 (Dodecahedral counterexample). *There is an origin-symmetric convex body $K \subset \mathbb{R}^3$ which is tangential to the Euclidean unit ball and satisfies*

$$\Gamma_{-2}K = B_2^3, \quad \Gamma_{-4}K = \left(\frac{5}{3}\right)^{1/4} B_2^3,$$

but is not an ellipsoid. Consequently, the property in Theorem 4.1(3) of the source does not characterize ellipsoids in $\mathcal{K}_e$.

3 Proof

Let $\varphi = (1 + \sqrt{5})/2$ and $q = \sqrt{1 + \varphi^2}$. Define

$$U = \frac{1}{q} \{(0, \pm 1, \pm \varphi), (\pm 1, \pm \varphi, 0), (\pm \varphi, 0, \pm 1)\} \subset S^2,$$

where the signs in each family are independent, giving twelve points. These are the vertices of a regular icosahedron. Put

$$K = \{x \in \mathbb{R}^3 : \langle x, u \rangle \leq 1 \text{ for every } u \in U\}.$$

Thus K is the polar regular dodecahedron. Since $U = -U$, the body is origin-symmetric. Each of its twelve facet planes is tangent to B_2^3 at its normal u , so K is tangential to the unit ball. The icosahedral symmetry is transitive on the facets, hence every facet has a common area $a > 0$.

We next calculate the required moments. For $x = (x_1, x_2, x_3)$, summing over the four independent sign choices in each coordinate family gives

$$\sum_{u \in U} \langle x, u \rangle^2 = 4|x|^2. \quad (3)$$

For the fourth moment, the same sign cancellation yields

$$\sum_{u \in U} \langle x, u \rangle^4 = \frac{4(1 + \varphi^4) \sum_i x_i^4 + 24\varphi^2 \sum_{i < j} x_i^2 x_j^2}{(1 + \varphi^2)^2}.$$

The identities $1 + \varphi^4 = 3\varphi^2$ and $(1 + \varphi^2)^2 = 5\varphi^2$ reduce this to

$$\sum_{u \in U} \langle x, u \rangle^4 = \frac{12}{5} \left(\sum_i x_i^2 \right)^2 = \frac{12}{5} |x|^4. \quad (4)$$

For this polytope,

$$dS_K = a \sum_{u \in U} \delta_u.$$

Moreover $h_K(u) = 1$ at every atom, so $dS_p(K) = dS_K$ in the integrals in (1). Decomposing K into the twelve pyramids with apex at the origin, base area a , and height one gives

$$V(K) = \frac{1}{3}(12a) = 4a. \quad (5)$$

Combining (1), (3), and (5),

$$\|x\|_{\Gamma_{-2}K}^2 = \frac{a}{4a} 4|x|^2 = |x|^2,$$

so $\Gamma_{-2}K = B_2^3$. Likewise (4) gives

$$\|x\|_{\Gamma_{-4}K}^4 = \frac{a}{4a} \frac{12}{5} |x|^4 = \frac{3}{5} |x|^4,$$

and therefore $\Gamma_{-4}K = (5/3)^{1/4} B_2^3$. Since $n = 3$, this is exactly (2). Finally, K has planar facets and is a polytope, whereas an ellipsoid is strictly convex and has no nontrivial boundary segments. Hence K is not an ellipsoid, proving the theorem.

4 Exact verification and novelty check

From this packet directory, run

```
conda run --no-capture-output -n sandbox python \
  code/verify_icosahedral_moments.py
```

The script expands both moment polynomials exactly over $\mathbb{Q}(\sqrt{5})$ and asserts (3)–(4); no floating-point decision is used.

A bounded search on 19 July 2026 covered the run’s lightweight indexes and local source corpus, together with exact web combinations of the source arXiv id, “ Γ_{-4} ,” “tangential body,” “dodecahedron,” and “spherical 4-design.” It found no later answer or this example. Novelty confidence is moderate–high, subject to a full bibliographic review.

References

- [1] E. Milman, *A sharp centro-affine isospectral inequality of Szegő–Weinberger type and the L^p -Minkowski problem*, [arXiv:2103.02994](https://arxiv.org/abs/2103.02994); J. Differential Geom. 127 (2024), 651–714.